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ORIGINAL ARTICLE

# Force enhancement during and following muscle stretch of maximal voluntarily activated human quadriceps femoris

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**Abstract** Force enhancement during and following muscle stretch has been observed for electrically and voluntarily activated human muscle. However, especially for voluntary contractions, the latter observation has only been made for adductor pollicis and the ankle joint muscles, but not for large muscles like quadriceps femoris. Therefore, the aim of this study was to investigate the effects of active muscle stretch on force production for maximal voluntary contractions of in vivo human quadriceps femoris ($n = 15$). Peak torques during and torques at the end of stretch, torques following stretch, and passive torques following muscle deactivation were compared to the isometric torques at corresponding muscle length. In addition, muscle activation of rectus femoris, vastus medialis and vastus lateralis was obtained using surface EMG. Stretches with different amplitudes (15, 25 and 35° at a velocity of 60° $s^{-1}$) were performed on the plateau region and the descending limb of the force–length relation in a random order. Data analysis showed four main results: (1) peak torques did not occur at the end of the stretch, but torques at the end of the stretch exceeded the corresponding isometric torque; (2) there was no significant force enhancement following muscle stretch, but a small significant passive force enhancement persisted for all stretch conditions; (3) forces during and following stretch were independent of stretch amplitude; (4) muscle activation during and following muscle stretch was significantly reduced. In conclusion, although our results showed passive force enhancement, we could not provide direct evidence that there is active force enhancement in voluntarily activated human quadriceps femoris.

**Keywords** Force enhancement · Eccentric muscle action · History of muscle action · Quadriceps femoris · Maximal voluntary activation

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## Introduction

When an electrically stimulated muscle is stretched, its force will exceed the isometric force at the corresponding muscle length (e.g., Fenn and Marsh 1935; Hill 1938). This phenomenon is known as force enhancement during stretch and is qualitatively well explained by Huxley's cross-bridge theory (Huxley 1957; Huxley and Simmons 1971); whereas, indicated by varying experimental data (e.g., Joyce et al. 1969; Edman et al. 1978; Linari et al. 2004), the quantitative description of force enhancement during stretch by means of cross-bridge models remains difficult (Harry et al. 1990; Herzog and Ait-Haddou 2003). However, during stretch of maximal voluntarily activated human muscles, eccentric forces are typically found to be similar to the forces of corresponding maximal isometric contractions (Westing et al. 1988; Westing et al. 1991; Webber and Kriellaars 1997; Babault et al. 2001; Reeves and Narici 2003), although there have also been contrary findings (Lee and Herzog 2002; Finni et al. 2003). The characteristics of eccentric forces during voluntary contractions are typically associated with neural inhibition (Webber and Kriellaars 1997; Babault et al. 2001) and more recently also with effective fascicle length changes, as observed by ultrasound (Reeves and Narici 2003).

In addition to force enhancement during stretch, there is substantial evidence that the steady-state isometric force

following stretch of electrically stimulated muscles remains greater than the corresponding force at the same muscle length obtained under purely isometric conditions (e.g., Edman et al. 1978; 1982; Herzog 1998; Linari et al. 2000). Furthermore, on the descending limb of the force–length relationship, the passive force following muscle stretch after deactivation was observed to exceed the passive force following an isometric reference contraction (Rassier et al. 2003; Herzog and Leonard 2005). These characteristics of the skeletal muscle are called active and passive force enhancement following stretch, respectively. As these properties cannot be explained in terms of the cross-bridge model and the degree of myofilament overlap (Gordon et al. 1966), several mechanisms have been suggested to be responsible for the experimental findings on force enhancement. The primary explanation has been the development of sarcomere length non-uniformities (e.g., Morgan 1990). But due to several reasons that have been tested experimentally (Peterson et al. 2004; Schachar et al. 2004; Herzog and Leonard 2005), this mechanism cannot account solely for the total active and passive force enhancement following muscle stretch. Thus, other mechanisms have been proposed to explain active and passive force enhancement. According to this, active force enhancement is speculated to be caused by an increase in the number of attached cross-bridges, a hypothesis that is supported by experiments that found an increased stiffness (Linari et al. 2000; Herzog and Leonard 2002) and a decrease in the rate of cross-bridge detachment following stretch (Herzog and Leonard 2002). Besides this, Pinniger et al. (2006) propose that the residual force enhancement is due to changes upon activation in parallel elastic elements. The passive component of force enhancement has also been associated with passive elements (Bagni et al. 2002) and recent findings suggest that the molecular spring titin, which changes its stiffness in a calcium-dependent manner, is the structure that is responsible (Labeit et al. 2003).

Besides this observation on the molecular level, if force enhancement following muscle stretch is to play a role in everyday movements, it has to occur in voluntarily activated human muscles (Lee and Herzog 2002). In their study, Lee and Herzog (2002) found that maximal voluntarily activated m. adductor pollicis showed active and passive force enhancement that was similar to force enhancement observed for electrically stimulated muscles. Moreover, it was shown that force enhancement is also present during submaximal voluntary contractions of m. adductor pollicis (Oskouei and Herzog 2005) as well as for submaximal plantar and dorsiflexion actions in humans (Pinniger et al. 2006). From these results, the authors concluded that force enhancement occurs in normal everyday movements so that it should have to be accounted for in biomechanical models of human movement. However, due to incomplete knowledge of the existence and the characteristics of force enhancement and its underlying mechanisms, especially in voluntarily activated large human muscles like the quadriceps group, its relevance in sports performance and daily living remains unclear.

Thus, in this study, we wanted to address the following questions: (1) Does force enhancement during and following muscle stretch exist in large human muscles that are voluntarily activated? (2) Do forces following active stretch depend on stretching amplitude, as reported in literature (Lee and Herzog 2002)? (3) How does muscle activation change during and following muscle stretch of voluntarily activated quadriceps femoris?

## Methods

### Subjects

Moderately active male subjects ($n = 15$; 22–30 years of age) with no history of neuromuscular disorder or injury at the knee joint of the right leg participated in this study. Free, written informed consent was obtained from all subjects and the study was conducted according to the Declaration of Helsinki.

### Torque and angle measurements

Knee torque, knee angle and angular displacement were measured by a rotational, isokinetic dynamometer (Model IsoMed2000, D&R Ferstl GmbH, Hemau, Germany). Data sampling frequency was 2,000 Hz.

### Muscle activation measurements

Muscle activation of m. rectus femoris, m. vast. medialis and m. vast. lateralis was obtained using surface electromyography (EMG). Recordings were made by means of bipolar surface electrodes with an inter-electrode distance of about 2 cm (AMBU® Blue Sensor R, Germany) placed on the relevant muscles. A single reference electrode (AMBU® Blue Sensor R, Germany) was attached to the patella. EMG signals were amplified no further than 15 cm from the recording site, and they were bandpass filtered (10 Hz and 1 kHz). EMG signals were collected at a sampling rate of 2,000 Hz. All measurements were synchronized by means of a software.

### Experimental settings

Subjects were placed on the dynamometer, with the horizontal seat reclined to 5° and the vertical backrest reclined to 15°, resulting in an angle of 100° between the seat and

the backrest. In order to restrict movement of the subjects, the pelvis was secured by a safety belt and the upper body by two safety belts and two additional shoulder pads. The shank was connected to the lever arm of the dynamometer, by two V-shaped pads that fit the anatomy of the tibia, and secured by two velcro straps. The center of the grapping torque element was aligned with the knee joint axis when subjects performed a maximal voluntary contraction.

## Experimental protocols

The investigation was based on the idea of examining the effects of different stretch amplitudes (15, 25 and 35° at a constant velocity of 60° $s^{-1}$) on force enhancement during and following muscle stretch. Therefore, the net knee joint moments, joint angles and muscle activation were measured for purely isometric reference contractions, for eccentric contractions during stretch and for isometric contractions following stretching of the quadriceps group. As we tried to reproduce the former findings on maximal voluntarily activated human muscles, the experimental protocol was similar to that of Lee and Herzog (2002).

Subjects attended three sessions on 3 different days, with at least 1 rest day between sessions. In two preparation sessions, subjects were familiarized with the test protocol and trained to perform maximal voluntary contractions. Subjects were instructed to develop their maximum force and then to maintain this force for the duration of the test. The start and end of each contraction was clearly announced to each subject and verbal encouragement was given during the contractions. In the third test session, subjects performed the actual test protocol.

The test protocol was split up into two parts. In part 1, subjects performed purely isometric contractions at 60, 70, 80 and 95° of knee flexion, whereby 0° knee flexion refers to the straight leg. For each knee angle, subjects had to make 3 repetitions, resulting in a total of 12 contractions. In part 2, for the determination of possible force enhancement during and following muscle stretch, subjects performed an isometric contraction at the relevant initial knee angle, followed by a stretch of m. quadriceps femoris, followed by a further isometric contraction at the final knee angle. The eccentric contractions started automatically, when subjects reached 95% of the torque that was realized during the maximal voluntary isometric reference contraction. Stretching was performed always at a constant angular velocity of 60° $s^{-1}$ over different stretch amplitudes of 15, 25 and 35° to the final knee angle of 95°. As the stretches started at 60, 70, and 80° of knee flexion, and the optimal length of the quadriceps group was expected to occur around 60° of knee flexion (Pincivero et al. 2004), the quadriceps group was stretched on the plateau region and the descending limb of the force–length relation. This was also confirmed by our results on the force–length relation (*F–l-r*) of quadriceps femoris (Fig. 1). Following stretching, the quadriceps muscle was kept activated for a minimum of 3 s to ensure that a steady-state force was reached following the stretch-transient force response. Similarly, following deactivation, the passive moment of the quadriceps group was measured for a minimum of 3 s at the final knee angle to determine possible force enhancement of passive structures. Each isometric–eccentric–isometric contraction of a given stretch magnitude was performed three times for a total of nine stretch experiments. In addition, to distinguish between total and active torques, purely passive torques were measured for isometric and stretch contractions (15, 25 and 35° stretch at a constant velocity of 60° $s^{-1}$) of the deactivated muscle and subsequently subtracted from the total torque resulting from the active trials.

In order to avoid possible learning or sequence effects during the experimental protocol, isometric contractions at different knee angles in part 1, as well as the different stretch amplitudes in part 2, were presented in a random order. Furthermore, in the isometric–eccentric–isometric contractions, the stretches only started automatically if subjects reached at least 95% of their maximum isometric moment at the initial knee angle. Subjects were given as much rest as required, but a minimum rest of 3 min was enforced at all times.

## Data analysis

Net knee joint moments were determined at six specific instances in time: peak moment during stretch ($M_{peak}$) and the eccentric moment at the end of the stretch ($M_{ecc}$); at 0.5–1 s (i.e., over a 500 ms interval), 1.5–2 s and 2.5–3 s

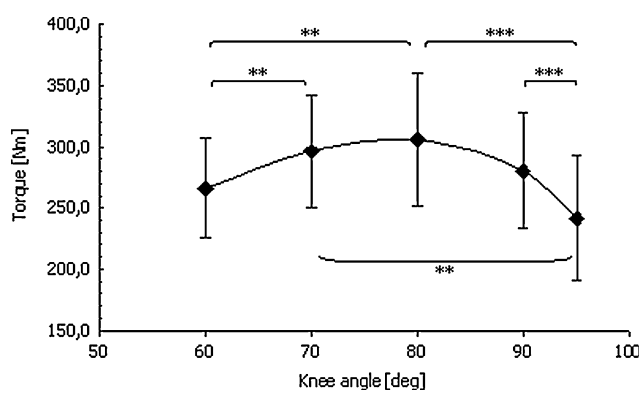


**Fig. 1** Moment–length relation of quadriceps femoris. Isometric torque was pooled for each knee angle and across all subjects. In general, optimal knee angle occurred around 80° knee flexion. Significance (** $P < 0.01$ and *** $P < 0.001$) is presented for differences in mean isometric torque for distinct knee flexion angles. According to the presented moment–length properties, stretches were mainly performed on the plateau region and the descending limb of quadriceps femoris

following stretch ($M_{iso}1$, $M_{iso}2$, and $M_{iso}3$) to give the mean steady-state moment following stretching, and at 1.5–2 s (i.e., over a 500 ms interval) after the termination of muscle activation ($M_{passive}$) to give the mean passive moment following deactivation of the quadriceps group. The $M_{ecc}$, $M_{iso}1$, $M_{iso}2$, and $M_{iso}3$ values were normalized relative to the corresponding mean moment (i.e., over a 500 ms interval) of the isometric reference contractions that were performed in part 1 of the experimental protocol. Force enhancement following muscle stretch (*FE*1, *FE*2 and *FE*3) was determined by subtracting the mean moments of the isometric reference contractions from the corresponding moments following active muscle stretching ($M_{iso}1$, $M_{iso}2$ and $M_{iso}3$). The $M_{passive}$ values were compared to the corresponding passive moments following the isometric reference contractions. For optimal comparability, all contractions were of similar duration and comparisons were done for equal knee angles and at the same point of time.

For the analysis of muscle activation, mean absolute values (MAV) of the root mean square (RMS) values of the EMG signals (for a 500 ms period) were calculated for the isometric reference contractions, during the stretch and during the isometric phase following stretch. The MAV of the EMG signals from the isometric–eccentric–isometric contractions were compared to the corresponding mean values of the isometric reference contractions (Fig. 2). For the calculation of correlations between *FE* and muscle activation, mean *FE* (i.e., average of *FE*1, *FE*2 and *FE*3) was correlated to mean muscle activation following the stretch of mm. vastus lateralis, vastus medialis, and rectus femoris, respectively.

Mean values of the measured variables for each testing condition across all subjects were used for statistical analysis ($\alpha = 0.05$). Repeated measures ANOVA and Bonferroni post hoc comparisons served for statistical analysis of isometric forces, forces during and following stretch and EMG signals. Pearson correlation was calculated between *FE* and the corresponding muscle activation. In the figures, the variables are presented as means ± 1 standard deviation.

## Results

For all stretching amplitudes, we found force enhancement during the stretch of voluntarily activated quadriceps femoris, but no significant active steady-state force enhancement following muscle stretch, although there was a small significant passive force enhancement after deactivation of the quadriceps group. Eccentric moments during stretch as well as active and passive forces following muscle stretching were independent of stretch amplitude. For all muscles and each stretch condition, muscle activation during and following stretch was smaller than during maximal voluntary isometric reference contractions.

### Knee joint moments during stretch

The peak moments during stretch ($M_{peak}$) did not occur at the end of the stretch. Moreover, the angular displacement to reach $M_{peak}$ was related to the amount of stretch, so that $M_{peak}$ occurred at different knee angles. The mean knee flexion angles for $M_{peak}$ to occur were 86.3 ± 1.5,

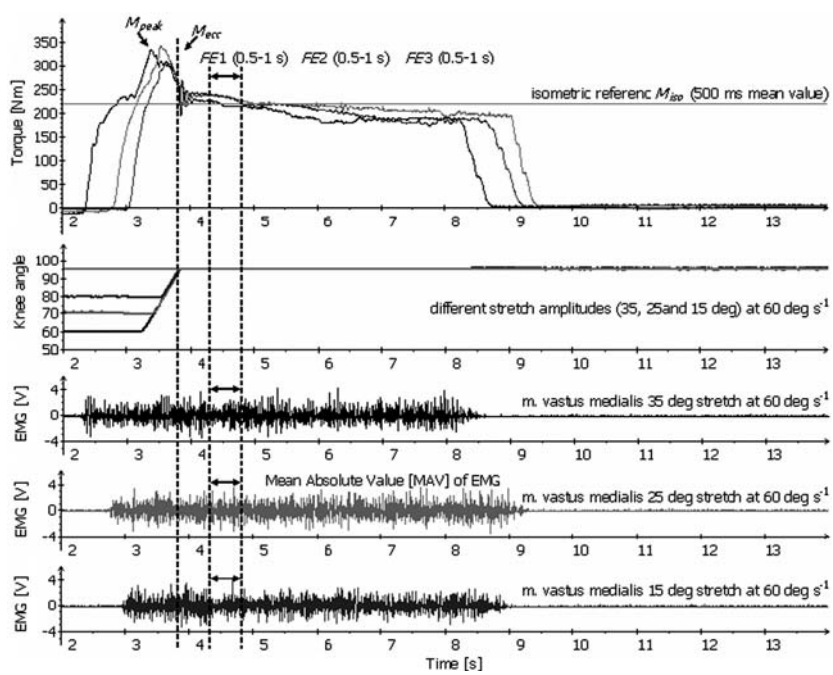


**Fig. 2** Knee joint torque and angle–time histories and EMG signals of isometric–eccentric–isometric test contractions obtained for part 2 of the test protocol. For data analysis, peak torque ($M_{peak}$) and eccentric torque ($M_{ecc}$) were measured during and at the end of the stretch, respectively, and isometric torques following stretch were determined at three instances of time. All parameters served for comparison with the corresponding isometric reference contraction and were used to assess force enhancement (*FE*). Also, mean absolute value of EMG was obtained before, during and after the stretch for determination of muscle activation (EMG of m. vastus medialis is presented)

78.9 ± 3.7 and 74.5 ± 8.9° after an angular displacement of 6.3 ± 1.5, 8.9 ± 3.7 and 14.5 ± 8.9° for the 15, 25 and 35° stretch, respectively. Although $M_{peak}$ appeared at distinct knee angles, it was independent of stretch amplitude. The mean $M_{peak}$ values were 337 ± 53, 347 ± 54 and 340 ± 58 Nm for the 15, 25 and 35° stretch, respectively.

The eccentric moments at the end of the stretch ($M_{ecc}$) exceeded the corresponding isometric torque ($P < 0.05$) for all stretching conditions and were independent of stretch amplitude. The mean $M_{ecc}$ values were 116 ± 14, 120 ± 15 and 118 ± 14% for the 15, 25 and 35° stretch, respectively.

### Isometric knee joint moments following muscle stretch

None of the isometric moments following stretching ($M_{iso}1$, $M_{iso}2$, and $M_{iso}3$) was significantly greater than the corresponding isometric reference moment for all stretch conditions. The mean "force enhancement" following the 15, 25, and 35° stretches ranged from 3 to 0% following muscle stretch (*FE*1), from −2 to −5%, 1.5–2 s following muscle stretch (*FE*2), and −7 to −11%, 2.5–3 s following muscle stretch (*FE*3), respectively, as shown in Fig. 3a. Statistics showed that *FE*1 was significantly greater than *FE*2 and *FE*3 ($P < 0.05$ for the 15, 25 and 35° stretch, respectively) for all stretching amplitudes. Similarly, *FE*2 exceeded *FE*3 ($P < 0.05$) for all stretch conditions (Fig. 3a). When comparing stretch conditions, there were no significant differences for *FE*1, *FE*2 and *FE*3, respectively. According to this, forces following stretch were independent of the stretch amplitude (Fig. 3b).

### Passive knee joint moment after the contraction ($M_{passive}$)

The passive moments following muscle stretching were significantly greater than the passive moments following the purely isometric reference contractions. This increase in passive torque is called "passive force enhancement" and was observed consistently for all stretch conditions. Moreover, passive force enhancement ($FE_{passive}$) was found to be independent of stretch amplitude (Fig. 4).

### Muscle activation during and following stretch

Regardless of the stretch amplitudes, EMG signals during and following active muscle stretching were smaller than those during the isometric reference contraction. In addition, for all muscles there was no systematic change in muscle activation as a function of stretch conditions. Thus, in comparison to maximal voluntary activation, the mean activation of the muscles rectus femoris, vastus medialis and vastus lateralis during and following stretch was 82 ± 15, 86 ± 13 and 83 ± 19%, respectively (Fig. 5). Moreover, we found moderate correlations between forces following stretch and muscle activation of mm. vastus lateralis ($r = 0.29$, $P = 0.026$), vastus medialis ($r = 0.22$, $P = 0.075$) and rectus femoris ($r = 0.34$, $P = 0.01$), respectively.

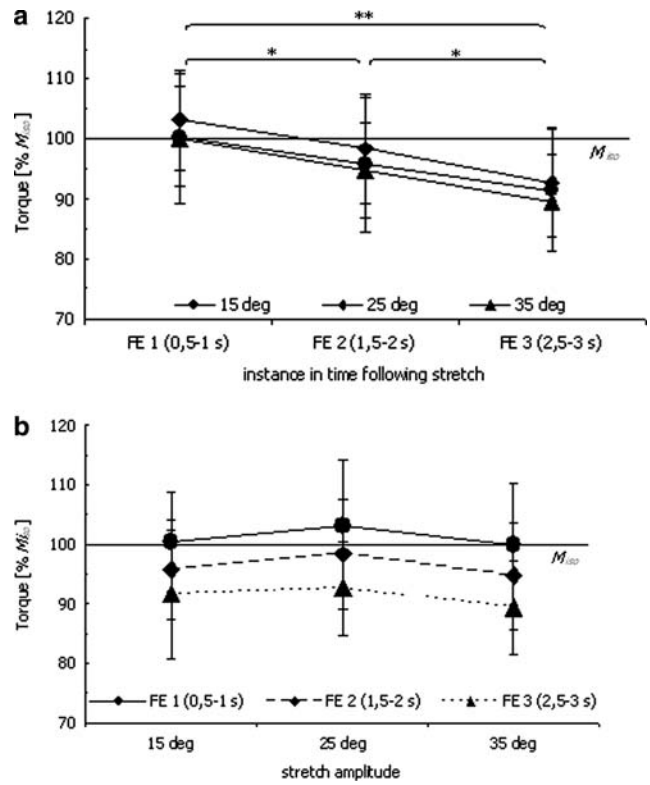


**Fig. 3** Force enhancement (*FE*) following muscle stretch. Force enhancement values (*FE*) were pooled for each stretch condition and across all subjects, and torques were normalized relative to the corresponding isometric reference torque ($M_{iso}$). In general, when compared to the isometric reference contraction, and for all stretch amplitudes, none of the isometric torques following muscle stretch (*FE*1, *FE*2 and *FE*3) showed any significant force enhancement. **a** For all stretch conditions, *FE*1 was significantly greater than *FE*2 (* $P < 0.05$) and *FE*3 (** $P < 0.01$), and *FE*2 was significantly greater than *FE*3 (*$P < .05$), whereby *p*-values only indicate the weakest significances. **b** Active forces following muscle stretch were independent of stretch amplitude

## Discussion

The purpose of this study was to investigate force enhancement during and following stretching of human quadriceps femoris during maximal voluntary activation. New to literature is the observation that the peak torques during stretch did not occur at the end of the stretch. In addition, the eccentric knee joint moments at the end of the stretch exceeded the torque of the isometric reference contraction. In contrast to former findings, there was no steady-state force enhancement following muscle stretching for voluntary contractions, although there was some passive force enhancement. Moreover, eccentric moments at the end of

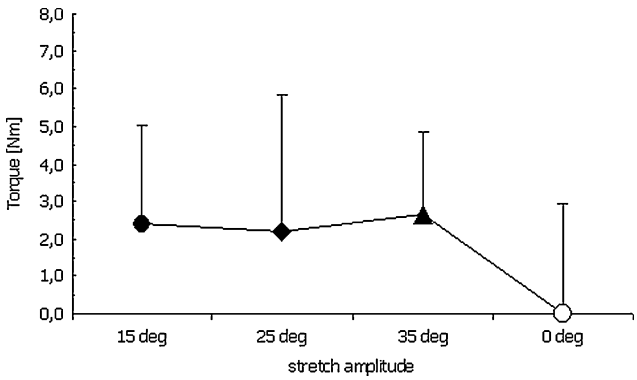


**Fig. 4** Passive force enhancement ($FE_{passive}$) after deactivation of the muscle. Passive torque values were pooled for each stretch condition and across all subjects. $FE_{passive}$ was calculated by subtracting the passive torque of the isometric reference contraction ($M_{iso}$) from the $M_{passive}$ values following deactivation after the isometric–eccentric–isometric test contractions. For presentation, $M_{iso}$ was set to 0 so that the *Y*-axis gives the difference between $M_{passive}$ and $M_{iso}$. For all stretch amplitudes, passive torques following active muscle stretching were significantly greater ($P < 0.05$) than the passive torque following the isometric reference contraction (0°). There was no significant difference due to the stretching amplitudes

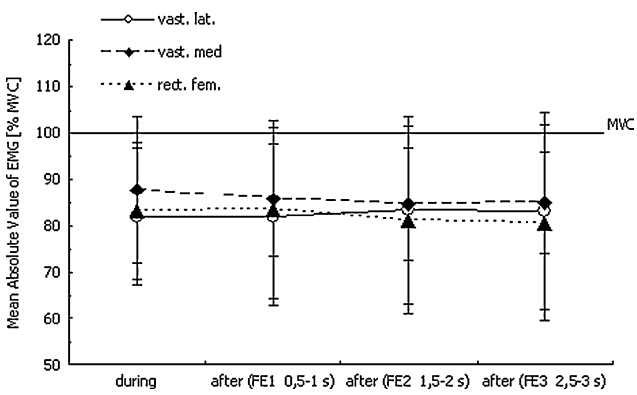


**Fig. 5** Muscle activation determined with EMG during and following active muscle stretch. The EMG values during and following stretch were pooled for all stretch conditions and across all subjects and normalized relative to activation during maximum voluntary contraction (MVC). For all muscles, activation during and following active muscle stretching was significantly ($P < 0.05$) smaller than that during the isometric reference contractions, as indicated by the mean absolute value (MAV) of the EMG. During: MAV of EMG just during active stretch for 500 ms; after: MAV of EMG 0.5–1 s, 1.5–2 s, and 2.5–3 s after active stretch for 500 ms during the second isometric phase of the isometric–eccentric–isometric test contraction

the stretch as well as active forces following stretch and passive force enhancement were independent of stretch amplitude.

## Force enhancement during stretch ($M_{peak}$, $M_{ecc}$)

Normally, peak eccentric forces for electrically as well as for voluntarily activated human muscles occur at the end of stretching (Webber and Kriellaars 1997; De Ruiter et al. 2000; Lee and Herzog 2002). Our results for the voluntarily activated quadriceps femoris do not support these published findings. Peak knee joint moments during stretch, averaged across all conditions, occurred after $9.9 \pm 4.3°$ of stretch, and the angular displacement to reach $M_{peak}$ was directly related to the amount of the stretch. That means, the shorter the stretch (i.e., the stretch started at a longer muscle length), the earlier is the appearance of $M_{peak}$. In addition, $M_{peak}$ occurred at different knee joint angles, but showed similar amounts of force.

Both these results might be explained by the *F–l-r* of quadriceps femoris, as obtained for our subjects (Fig. 1). For each stretch condition, $M_{peak}$ was found at the plateau region of the *F–l-r* at an optimal knee angle, which in contrast to Pincivero et al. (2004) occurred between 75 and 85° of knee flexion, respectively. Since our stretches started at different knee angles (60, 70 and 80°, respectively), the displacement to reach (for the stretches starting at 60 and 70°, respectively) or to stay in (the stretch starting at 80°) the plateau region varied between each stretch condition. Thus, it suggests itself that the amount of $M_{peak}$ did not differ among stretch conditions as all of them occurred around an optimal knee angle.

In agreement with the results of Lee and Herzog (2002), but in contrast to former findings concerning the knee joint (Westing et al. 1988; 1991; Webber and Kriellaars 1997), in our study, eccentric moments at the end of the stretch ($M_{ecc}$) exceeded the corresponding isometric knee joint moment. There are two possible explanations to account for this difference: (1) In contrast to Westing et al. (1988; 1991) who started their stretches at a certain low force level, our stretches were preceded by a maximum voluntary isometric contraction. (2) In comparison to Webber and Kriellaars (1997), the stretch amplitudes differed strongly, so that due to less amount of stretching in the present study neural muscle inhibition (Westing et al. 1988; Dudley et al. 1990; Webber and Kriellaars 1997; Babault et al. 2001) may play a smaller role in relatively short stretches compared to stretches over a large range of motion (e.g., a 90° stretch in the study of Webber and Kriellaars 1997).

## Active forces enhancement following stretch (*FE1, FE2, FE3*) and passive knee joint moment after deactivation ($M_{passive}$)

Although steady-state force enhancement following stretching of voluntarily activated muscles has been demonstrated for m. adductor pollicis and the muscles crossing the ankle joint (Lee and Herzog 2002; Oskouei and Herzog 2005; Pinniger et al. 2006), the question arises if there also is active force enhancement in maximal voluntarily activated, large human muscles like m. quadriceps femoris.

Our results did not show any significant active force enhancement following stretching of quadriceps femoris muscle. Moreover, the characteristics of post eccentric moments (e.g., the independency on stretch amplitude) were found to be in contrast to the former results (Lee and Herzog 2002; Herzog et al. 2006) that reported an optimal stretch amplitude of 20° for voluntarily activated human adductor pollicis and an increase of force enhancement with increase of stretching amplitude for electrically stimulated muscles.

However, in agreement with Lee and Herzog (2002), we provided direct evidence for a small, but significant, passive force enhancement that was independent of stretch amplitude. According to this, the problem that there was some passive, but no active, force enhancement arises. To our knowledge, such a result has not been reported in literature, and it seems quite unlikely that there was passive but no active force enhancement. Thus, we would like to present some aspects of our results that may help to explain our observations.

According to their observations on force enhancement in submaximal voluntary contractions, Oskouei and Herzog (2005) proposed that force enhancement is defined as a decrease in steady-state isometric EMG in the isometric–eccentric–isometric test contraction compared with the EMG obtained in the purely isometric reference contraction. That means that if forces following stretching are higher or equal to the corresponding isometric forces while activation is reduced, there is active force enhancement, as the same force is achieved with lower activation. As our results showed, in comparison to EMG during maximal voluntary isometric contraction, muscle activation following muscle stretching was significantly reduced to about 14–18%. To that effect and in accordance with the suggestion of Oskouei and Herzog (2005), our results indicate that there was active force enhancement in maximal voluntarily activated m. quadriceps femoris, 0.5–1 s (*FE1*) following stretch for all stretch amplitudes. This is due to the finding that 0.5–1 s after stretch, the torque reached 100% of the corresponding isometric reference contraction. However, since all other knee joint moments following stretching dropped below the level of the isometric reference contraction, it is difficult to separate the activation-induced decrease in force.

In order to test the Oskouei and Herzog hypothesis further, we investigated the correlation between forces following stretch and muscle activation. If their suggestion applies, forces following stretch and muscle activation should show a highly positive correlation. In testing this, we only found moderate correlations between forces following stretch and muscle activation of mm. vastus lateralis ($r = 0.29$, $P = 0.026$), vastus medialis ($r = 0.22$, $P = 0.075$), and rectus femoris ($r = 0.34$, $P = 0.01$), respectively.

However, despite the precautions set in the section "Methods" to avoid a sequence effect, such an effect could be another factor that may contribute to our finding that there was no obvious force enhancement following stretch. Since EMG was found to be consistently lower during the isometric phase following stretch, we cannot exclude that some central fatigue had occurred following the eccentric contractions.

In addition to the activation-inhibition problem, Oskouei and Herzog (2005) observed that force enhancement was only apparent in about one-half of their tested subjects. In accordance with this observation, we also found 4 out of 15 tested subjects that had a consistent force enhancement of $14 \pm 4\%$ (*FE1*), $6 \pm 6\%$ (*FE2*) and $1 \pm 8\%$ (*FE3*) averaged across all stretching conditions and 6 out of 15 subjects that showed a force enhancement of $11 \pm 6\%$ (*FE1*), $5 \pm 7\%$ (*FE2*) and $0 \pm 9\%$ (*FE3*) for the 15 and 25° stretch, respectively. According to the proposal of Oskouei and Herzog (2005) concerning muscle activation and force enhancement, it should be expected that the four subjects, who showed force enhancement also had a higher muscle activation. By means of statistical analysis (Student's *t*-test), we found that activation of m. rectus femoris differed significantly ($P < 0.05$) between the groups that showed force enhancement ($85 \pm 10\%$) and the ones that did not ($78 \pm 15\%$), whereas vastus lateralis ($85 \pm 26$ vs. $82 \pm 18\%$) and vastus medialis ($86 \pm 15$ vs. $84 \pm 12\%$) had similar activation levels.

Thus, due to the fact that we only found moderate correlations between forces following stretch and muscle activation as well as only a slight increase in muscle activation of m. rectus femoris, it seems quite likely that besides the activation-inhibition phenomenon, other factors contribute to influence post eccentric forces.

In this regard, as demonstrated by Oskouei and Herzog (2005), those subjects who showed force enhancement had greater postactivation potentiation and a smaller resistance to fatigue. From both these results, the authors conclude that the group showing force enhancement might have had a greater percentage of fast-twitch fibers than the group who showed no force enhancement. Additionally, it was also shown by others that eccentric forces during stretch depend on muscle fiber type (Mutungi and Ranatunga 2001; Linari et al. 2004). According to this, when analysing our data, we found that the eccentric forces at the end of the stretch ($M_{ecc}$) were strongly correlated ($r = 0.89$, $P = 0.01$) with the forces following stretch (*FE*). Based on these aspects, an additional factor that contributes to force enhancement during and following muscle stretch might therefore be the fiber type constitution of the tested muscle (Pinniger et al. 2006).

Altogether, due to the associated inhibitions when stretching large muscles in the human body such as the

quadriceps group (Babault et al. 2001; Beltman et al. 2004), the lack of force enhancement seems to be a combined activation-inhibition phenomenon. In consideration of the factors discussed above, we therefore suppose that the decrease in activation is the main factor for hiding the actual force enhancement. Thus, we also assume that we observed passive force enhancement because there was a hidden active force enhancement.

## Conclusion

We demonstrated that the eccentric torques during the stretch of voluntarily activated quadriceps muscle exceeded the corresponding isometric torque. These findings could be explained in terms of cross-bridge theory and force–length properties of the skeletal muscle, respectively. Although we did not find direct evidence for active force enhancement following stretch in maximal voluntary activated human quadriceps femoris, with respect of our results we presented some arguments that would support its existence: Based on our finding that passive force enhancement occurred in the quadriceps muscle, the question arose why there was no apparent active force enhancement. In this context, we suggest that there was active force enhancement that was mainly hidden by reduced activation and additional neural inhibition of quadriceps femoris muscle. Furthermore, force enhancement might also be related to other factors like fiber type constitution.

Recapitulating, we could not find direct evidence that the mechanisms of force enhancement play a role in everyday movements. On the other hand, we presented some aspects of our results which support the existence of force enhancement in maximal voluntarily activated human quadriceps femoris. However, as knowledge of the exact characteristics of force enhancement (e.g., sub-maximal activation levels, influence of fiber type, effective muscle length, high velocity stretches) is still incomplete, it remains difficult to account for force enhancement properties in analysis or in biomechanical models of human movement. Thus, there are still several ambiguous factors, which influence force enhancement of voluntarily activated human muscles that need to be addressed.

## References


Babault N, Pousson M, Ballay Y, Van Hoecke J (2001) Activation of human quadriceps femoris during isometric, concentric, and eccentric contractions. J Appl Physiol 91:2628–2634

Bagni MA, Cecchi G, Colombini B, Colomo F (2002) A non-cross-bridge stiffness in activated frog muscle fibers. Biophys J 82:3118–3127

Beltman JG, Sargeant AJ, van Mechelen W, de Haan A (2004) Voluntary activation level and muscle fiber recruitment of human quadriceps during lengthening contractions. J Appl Physiol 97:619–626

De Ruiter CJ, Didden WJ, Jones DA, Haan AD (2000) The force–velocity relationship of human adductor pollicis muscle during stretch and the effects of fatigue. J Physiol 526(Pt 3):671–681

Dudley GA, Harris RT, Duvoisin MR, Hather BM, Buchanan P (1990) Effect of voluntary vs. artificial activation on the relationship of muscle torque to speed. J Appl Physiol 69:2215–2221

Edman KA, Elzinga G, Noble MI (1978) Enhancement of mechanical performance by stretch during tetanic contractions of vertebrate skeletal muscle fibres. J Physiol 281:139–155

Edman KA, Elzinga G, Noble MI (1982) Residual force enhancement after stretch of contracting frog single muscle fibers. J Gen Physiol 80:769–784

Fenn WO, Marsh BS (1935) Muscular force at different speeds of shortening. J Physiol 58:373–395

Finni T, Ikegawa S, Lepola V, Komi PV (2003) Comparison of force–velocity relationships of vastus lateralis muscle in isokinetic and in stretch-shortening cycle exercises. Acta Physiol Scand 177:483–491

Gordon AM, Huxley AF, Julian FJ (1966) The variation in isometric tension with sarcomere length in vertebrate muscle fibres. J Physiol 184:170–192

Harry JD, Ward AW, Heglund NC, Morgan DL, McMahon TA (1990) Cross-bridge cycling theories cannot explain high-speed lengthening behavior in frog muscle. Biophys J 57:201–208

Herzog W (1998) History dependence of force production in skeletal muscle: a proposal for mechanisms. J Electromyogr Kinesiol 8:111–117

Herzog W, Ait-Haddou R (2003) Mechanical muscle models and their application to force and power production. In: Komi PV (ed) Strength and power in sport. Blackwell , Oxford, pp 154–187

Herzog W, Lee EJ, Rassier DE (2006) Residual force enhancement in skeletal muscle. J Physiol 574:635–642

Herzog W, Leonard TR (2002) Force enhancement following stretching of skeletal muscle: a new mechanism. J Exp Biol 205:1275–1283

Herzog W, Leonard TR (2005) The role of passive structures in force enhancement of skeletal muscles following active stretch. J Biomech 38:409–415

Hill AV (1938) The heat of shortening and the dynamic constants of muscle. Proceedings of the Royal Society of London 126:136–195

Huxley AF (1957) Muscle structure and theories of contraction. Prog Biophys Biophys Chem 7:255–318

Huxley AF, Simmons RM (1971) Proposed mechanism of force generation in striated muscle. Nature 233:533–538

Joyce GC, Rack PM, Westbury DR (1969) The mechanical properties of cat soleus muscle during controlled lengthening and shortening movements. J Physiol 204:461–474

Labeit D, Watanabe K, Witt C, Fujita H, Wu Y, Lahmers S, Funck T, Labeit S, Granzier H (2003) Calcium-dependent molecular spring elements in the giant protein titin. Proc Natl Acad Sci U S A 100:13716–13721

Lee HD, Herzog W (2002) Force enhancement following muscle stretch of electrically stimulated and voluntarily activated human adductor pollicis. J Physiol 545:321–330

Linari M, Bottinelli R, Pellegrino MA, Reconditi M, Reggiani C, Lombardi V (2004) The mechanism of the force response to stretch in human skinned muscle fibres with different myosin isoforms. J Physiol 554:335–352

Linari M, Lucii L, Reconditi M, Casoni ME, Amenitsch H, Bernstorff S, Piazzesi G, Lombardi V (2000) A combined mechanical and X-ray diffraction study of stretch potentiation in single frog muscle fibres. J Physiol 526(Pt 3):589–596

Morgan DL (1990) New insights into the behavior of muscle during active lengthening. Biophys J 57:209–221

Mutungi G, Ranatunga KW (2001) The effects of ramp stretches on active contractions in intact mammalian fast and slow muscle fibres. J Muscle Res Cell Motil 22:175–184

Oskouei AE, Herzog W (2005) Observations on force enhancement in submaximal voluntary contractions of human adductor pollicis muscle. J Appl Physiol 98:2087–2095

Peterson DR, Rassier DE, Herzog W (2004) Force enhancement in single skeletal muscle fibres on the ascending limb of the force–length relationship. J Exp Biol 207:2787–2791

Pincivero DM, Salfetnikov Y, Campy RM, Coelho AJ (2004) Angle- and gender-specific quadriceps femoris muscle recruitment and knee extensor torque. J Biomech 37:1689–1697

Pinniger GJ, Ranatunga KW, Offer GW (2006) Crossbridge and non-crossbridge contributions to tension in lengthening rat muscle: force-induced reversal of the power stroke. J Physiol 573:627–643

Rassier DE, Herzog W, Pollack GH (2003) Stretch-induced force enhancement and stability of skeletal muscle myofibrils. Adv Exp Med Biol 538:501–515

Reeves ND, Narici MV (2003) Behavior of human muscle fascicles during shortening and lengthening contractions in vivo. J Appl Physiol 95:1090–1096

Schachar R, Herzog W, Leonard TR (2004) The effects of muscle stretching and shortening on isometric forces on the descending limb of the force–length relationship. J Biomech 37:917–926

Webber S, Kriellaars D (1997) Neuromuscular factors contributing to in vivo eccentric moment generation. J Appl Physiol 83:40–45

Westing SH, Cresswell AG, Thorstensson A (1991) Muscle activation during maximal voluntary eccentric and concentric knee extension. Eur J Appl Physiol Occup Physiol 62:104–108

Westing SH, Seger JY, Karlson E, Ekblom B (1988) Eccentric and concentric torque-velocity characteristics of the quadriceps femoris in man. Eur J Appl Physiol Occup Physiol 58:100–104